

THE COMPLETE FRAMEWORK Organ Procurement Performance, Governance, and Accountability System (OPPGAS)

A De Novo Evaluation Architecture with Full Mathematical Specification

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Abstract

This document presents the complete, mathematically specified evaluation framework for organ procurement organization oversight. It includes mechanisms designed for immediate implementation, near-term rulemaking, and long-term architectural transformation. Related submissions: Comments 1–7 on this docket.

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1 I. Foundational Design Philosophy

This framework optimizes for a single terminal metric: organs successfully transplanted per eligible death, nationally. Every other metric in this document is instrumental to that objective. The framework is structurally indifferent to how many OPOs exist, what their boundaries are, or who operates them. It cares only that every recoverable organ reaches a recipient.

The framework rests on five architectural axioms, each derived from observed failure modes documented on this docket across Comments 1–7:

1.1 Axiom 1: Absolute Standards over Relative Rankings

The current tier system uses prior-year thresholds that partially mitigate forced-curve dynamics, and 30 of 55 OPOs achieved Tier 1 in 2025. This framework preserves that achievement while introducing risk-adjusted absolute standards that distinguish OPO performance from DSA difficulty with mathematical precision. If all 55 OPOs perform excellently, all 55 are certified. The number of survivors is determined by reality, not by percentile cutoffs.

1.2 Axiom 2: Output Metrics Are Necessary but Insufficient

Clinical output (donation rate, transplant rate) measures what comes out of the organization. It cannot detect the governance architecture that produces those outcomes. An OPO can meet benchmarks for years while its internal decision-making structure selects against competence, transparency, and honest self-assessment. The framework supplements output measurement with governance, efficiency, and process architecture that makes the Performance Layer structurally difficult to sustain.

1.3 Axiom 3: Gaming Resistance through Aligned Incentives

Every metric will be optimized against. The framework is designed so that the optimal gaming strategy is indistinguishable from genuine performance improvement. Where direct measurement can be fabricated, the framework measures the delta between narrative artifacts (controllable) and physics-bound exhaust (not controllable). When the cheapest way to appear compliant is to actually be compliant, the Performance Layer’s business model collapses.

1.4 Axiom 4: Financial Consequence over Reporting Requirements

Transparency does not equal accountability when the regulator is chronically under-resourced and the regulated entity is highly incentivized to obfuscate. Where possible, the framework links measurement directly to Medicare reimbursement mechanics, creating self-executing financial consequences that do not depend on CMS staff actively monitoring 55 quarterly dashboards. Dashboards serve the board and staff; automated financial triggers serve enforcement.

1.5 Axiom 5: The Clinical Workforce Is Never the Problem

Every mechanism in this framework protects the clinical core wire—coordinators, recovery teams, family services staff, surgical teams—while creating accountability for the administrative scaffolding above them. During any governance transition, the clinical workforce transfers intact. The institutional knowledge, hospital relationships, and community trust they carry are the irreplaceable assets. Everything above them is replaceable.

2 II. Core Performance Measurement Architecture

2.1 II.A. Recovery Efficiency Index (REI)

The REI replaces raw per-eligible-death ratios with a risk-adjusted measure that isolates OPO performance from DSA structural characteristics.

2.1.1 Definition

$$\text{REI}_i = \frac{O_i}{E[O_i | X_i]}$$

Where O_i is actual organs transplanted by OPO i and $E[O_i | X_i]$ is expected organs transplanted given DSA characteristic vector X_i . An REI of 1.0 means the OPO recovers exactly what the model predicts. Above 1.0: outperformance. Below 1.0: underperformance.

2.1.2 Risk Adjustment Model Specification

Expected yield is computed via negative binomial regression (to handle overdispersion in count data):

$$\ln(E[O_i]) = \alpha + \beta_1 \ln(D_i) + \beta_2 \ln(\rho_i) + \beta_3 T_i + \beta_4 R_i + \beta_5 G_i + \beta_6 \ln(I_i) + \beta_7 C_i + \varepsilon_i$$

Variable definitions:

- D_i = eligible deaths per 100,000 population (CDC MCODE data, already collected)
- ρ_i = population density of DSA (Census Bureau, determines transport feasibility)
- T_i = trauma center density per capita (AHA Annual Survey, determines referral quality)
- R_i = registered donor rate by state (Donate Life America, cultural baseline)
- G_i = geographic dispersion index (mean distance from donor hospitals to transplant centers, km)
- I_i = median household income of DSA (Census/ACS, proxy for healthcare access)
- C_i = transplant center count within DSA (SRTR, determines placement capacity)

Coefficients are estimated nationally using 3–5 years of historical data across all 55 DSAs. The model is published. The methodology is transparent. Any OPO can compute its own expected yield. Coefficients are updated each recertification cycle (4 years).

2.1.3 The Demographic Adjustment Question

Whether to include race, ethnicity, and income-weighted consent rate differentials presents a genuine design tension requiring explicit resolution.

Model A (Include demographics): Improves model fit by $R^2 \approx 0.08$ – 0.12 . Accounts for documented consent rate differentials across populations. Prevents penalizing OPOs serving communities with historically lower consent rates driven by systemic medical mistrust that predates the OPO’s existence.

Model B (Exclude demographics): Establishes a normative mandate: overcoming community medical mistrust is a core competency the OPO is franchised to execute. Including demographics in the

model grants statistical credit for failing to invest in culturally competent outreach. The framework should measure what the OPO should achieve, not what history predicts it will achieve.

Resolution: CMS should compute and publish both REI-A (demographics included) and REI-B (demographics excluded) during the shadow period. The gap between REI-A and REI-B for a given OPO quantifies the magnitude of the demographic headwind. Regulatory consequences should attach to REI-B (the harder standard) while REI-A informs resource allocation for CMS technical assistance.

2.1.4 Confidence Intervals

For OPOs with small DSAs (< 200 eligible deaths/year), the REI point estimate carries wide uncertainty. Following CMS’s existing approach, the framework uses one-sided 95% confidence intervals:

$$\text{REI}_{i,\text{lower}} = \frac{O_i}{E[O_i]} \times \exp(-1.645 \times \text{SE}(\ln(\text{REI}_i)))$$

Where SE is the standard error of the log-REI estimate. An OPO is flagged only when $\text{REI}_{i,\text{lower}} < \text{threshold}$, giving smaller DSAs statistical benefit of the doubt.

2.1.5 Certification Thresholds

REI Range	Status	Consequence
≥ 1.0	Full Certification	No intervention. 4-year cycle.
0.85–0.99	Monitoring	Enhanced reporting, CMS liaison, voluntary improvement plan.
0.70–0.84	Improvement Required	Mandatory PIP, independent consultant, Medicare cost disallowance on executive compensation.
< 0.70 (2 yrs)	Supervised Operation	CMS-appointed oversight manager. DSA opened to expressions of interest.
< 0.50 (1 yr)	Emergency	Immediate competitive bid process. CMS-appointed administrator.

2.2 II.B. Absolute Output Contribution (AOC)

An OPO can have a high REI (performing well relative to its DSA’s predicted yield) while contributing almost nothing to the national organ supply because the DSA itself is structurally limited. AOC captures this:

$$\text{AOC}_i = \frac{O_i}{\sum_j O_j}$$

Where the denominator is total national organs transplanted. AOC measures each OPO's share of the national organ supply.

2.2.1 Joint Classification Matrix

REI and AOC are evaluated jointly:

	AOC $\geq$ Floor	AOC $<$ Floor
REI ≥ 1.0	Optimal	Structural Review of DSA — High REI + Low AOC $\rightarrow$ DSA may be non-viable as standalone territory. Evaluate boundary redesignation.
REI < 0.85	OPO Underperformance	Both Failing — Low REI regardless of AOC $\rightarrow$ graduated intervention ladder applies.

The AOC floor should be set at $\frac{1}{2N}$ where $N = 55$ (number of OPOs), i.e., approximately 0.9%. An OPO contributing less than half the national average share, even with adequate risk-adjusted performance, triggers a structural review of whether the DSA should exist as a standalone territory.

3 III. Process Efficiency Framework

CMS measures output (donation rate, transplant rate) but not cost-per-output. The following metrics detect the financial signature of governance dysfunction: duplicated labor costs, administrative bloat, and executive compensation disconnected from performance.

3.1 III.A. Cost Per Transplanted Organ (CPTO)

$$\text{CPTO}_i = \frac{TC_i}{O_i}$$

Where TC_i is total allowable costs from the audited Medicare cost report, cross-validated against Form 990 total expenses. Significant discrepancy ($\Delta > 10\%$) between cost report and 990 triggers automatic MAC audit. CPTO is risk-adjusted for geographic dispersion:

$$\text{CPTO}_{i,\text{adj}} = \frac{\text{CPTO}_i}{1 + \gamma \cdot G_i}$$

Where γ is the elasticity of cost with respect to geographic dispersion, estimated from the national cross-section. An 89-county DSA has legitimately higher transport costs than a 5-county urban DSA; the adjustment accounts for this.

3.2 III.B. Clinical Labor Efficiency (CLE)

$$\text{CLE}_i = \frac{O_i}{\text{FTE}_{i,\text{clinical}}}$$

Organs transplanted per clinical FTE. Clinical FTE is defined by the new §486.326(e) licensure requirements with a physical deployment criterion: the individual must spend more than 50% of billed hours physically deployed to donor hospitals or in active, direct clinical charting. Personnel whose primary function is managing other personnel must be classified as administrative overhead regardless of underlying clinical licensure. This prevents title inflation—reclassifying managers as “clinical FTEs” by requiring them to maintain unused licenses.

3.3 III.C. Administrative Overhead Ratio (AOR)

$$\text{AOR}_i = \frac{\text{FTE}_{i,\text{admin}} \times \bar{W}_{i,\text{admin}}}{TC_i}$$

Non-clinical staff costs as a percentage of total costs. This detects the double-pay architecture—organizations where the administrative layer consumes a disproportionate share of resources relative to clinical output.

Automated Trigger: When AOR_i exceeds the risk-adjusted peer median by 15 percentage points for two consecutive years, the MAC automatically initiates a Targeted Financial Audit and freezes future rate-increase requests until the dependency architecture is resolved. This transforms the dashboard from a passive publication into an active enforcement trigger.

3.4 III.D. Executive Compensation Efficiency (ECE)

$$\text{ECE}_i = \frac{\sum \text{ExecComp}_i}{O_i}$$

Total executive compensation (all officers above \$200,000, from Form 990 Part VII, including deferred, in-kind, and related-party compensation) divided by organs transplanted. Published as dollars per organ alongside organizational performance trajectory. An OPO paying \$2M in executive compensation while producing 300 organs is paying \$6,667 per organ in executive overhead. Published annually alongside REI.

3.5 III.E. The Performance Layer Deficit (D_{true})

The complete formulation of organizational governance cost:

$$D_{\text{true}} = C_{\text{leadership}} + V_{\text{destroyed}} - V_{\text{liability absorbed}}$$

Where:

- $C_{\text{leadership}}$ = total cost of leadership layer (executive compensation + administrative overhead + consulting + legal + board costs)
- $V_{\text{destroyed}}$ = value actively destroyed by leadership actions (expelled competent employees $\times$ replacement cost, donor volume lost to consent-rate decline from institutional dysfunction, opportunity cost of merger theater, clinical output degraded by dependency architecture, vendor/partner confidence erosion)
- $V_{\text{liability absorbed}}$ = genuine liability absorption value (the degree to which leadership's presence reduces the organization's legal and regulatory exposure below what it would be under flat governance)

When $D_{\text{true}} > 0$, the leadership layer is net-negative. The organization would produce more output at lower cost if the leadership layer were replaced. The transition catalyst role is viable only against organizations where D_{true} is positive and large.

3.5.1 Empirical Estimation from Public Data

D_{true} cannot be directly observed externally. However, a proxy can be computed from public filings:

$$\hat{D}_{\text{true}} \approx (\text{LRR}_i - \widetilde{\text{LRR}}) \times \text{Rev}_i + (\text{ECE}_i - \widetilde{\text{ECE}}) \times O_i$$

Where $\widetilde{\text{LRR}}$ is the median labor-to-revenue ratio. This measures excess labor cost plus excess executive compensation relative to peer medians, scaled by organizational size. Case evidence on this docket documents a $\hat{D}_{\text{true}}$ of approximately \$12–17M at a Texas-based OPO with 57.2% LRR versus a 40–45% peer median, producing an estimated \$25–35M when $V_{\text{destroyed}}$ is included.

4 IV. Governance Quality Assessment System

Governance quality is inherently harder to measure than clinical output. Any checklist becomes part of the Performance Layer it is designed to detect. This system measures behavioral patterns and physics-bound signals, not policy existence.

Design Principle: Measure the signals the Performance Layer must suppress to survive.

4.1 IV.A. Information Flow Integrity

Requirement: All OPO internal reporting mechanisms (ethics hotlines, compliance reporting, quality concern channels) must be operated by an independent third party with no contractual, employment, or financial relationship with the OPO's executive leadership or IT department. Reports are received by the third party, anonymized, and forwarded simultaneously to three independent bodies: (1) the board's audit committee, (2) CMS, and (3) the HHS Office of Inspector General (OIG). The OPO's IT department has zero access to the reporting infrastructure, the identity of complainants, or the content of reports prior to receipt by all three bodies.

Jurisdictional Deadlock: Simultaneous routing to three independent enforcement nodes makes coordinated suppression structurally impossible. The board can retain counsel to investigate internally, but it cannot legally suppress an active OIG inquiry. CMS can defer, but OIG operates on independent statutory authority. The information exists in at least one channel the Performance Layer cannot reach.

Cost: \$15,000–\$30,000 annually per OPO (NAVEX, EthicsPoint standard rates). Total national cost: \$825,000–\$1,650,000. Less than two weeks of CEO compensation at the documented organization.

4.2 IV.B. Personnel Pattern Monitoring

Requirement: OPOs must report quarterly to CMS: all voluntary and involuntary separations by department, role, and tenure; all positions vacant for more than 90 days; the ratio of management/supervisory positions to direct clinical positions; and any department where annual turnover exceeds 40% on a rolling 12-month basis.

4.2.1 The Whistleblower-to-Termination Ratio (WTR)

Traditional retaliation detection requires depositions and intent evidence. The WTR provides a statistical heuristic that algorithmically fingerprints cover-ups:

$$\text{WTR}_{i,d} = \frac{W_{i,d} / E[W_i]}{S_{i,d} / E[S_i]}$$

Where $W_{i,d}$ is the number of whistleblower complaints filed in department d at OPO i within a 12-month period, $E[W]$ is the expected complaint rate given organizational size, $S_{i,d}$ is involuntary separations in the same department within 6 months following the complaints, and $E[S]$ is the expected separation rate.

When $\text{WTR} \gg 1.0$, the department is generating both complaints and terminations at rates significantly above organizational baselines. This is the statistical signature of retaliation: complaints

trigger terminations in the same unit. The ratio bypasses the intent question entirely—it measures the pattern, not the motive.

Intervention trigger: $WTR > 2.0$ for any department triggers automatic on-site CMS review. $WTR > 3.0$ triggers simultaneous OIG referral.

4.2.2 Supervisor Concentration Index (SCI)

$$SCI_{i,s} = \frac{InvolSep_{i,s}}{InvolSep_{i,total}}$$

The proportion of all involuntary separations at OPO i attributable to a single supervisor s . Case evidence documents a single supervisor outside the formal reporting chain executing a multi-year pattern of personnel eliminations. $SCI > 0.30$ (one supervisor responsible for more than 30% of all involuntary separations in any 12-month period) triggers review.

Cross-validation: Personnel reports are cross-validated against state unemployment insurance filings during recertification survey. Reclassifying involuntary separations as “mutual” or “voluntary” is detectable when the separated employee files for unemployment benefits.

4.3 IV.C. Quarterly Financial Transparency Dashboard

Requirement: OPOs must submit quarterly to CMS and publish simultaneously to the board and all staff: total cash position broken into encumbered (pledged against debt instruments, restricted by grants, committed by contracts) and unencumbered categories; all outstanding debt instruments with covenant thresholds and current compliance status; current DSCR calculation with trailing 12-month trend; labor-to-revenue ratio trended over eight quarters; and executive compensation as approved by the board compared to organizational performance trajectory.

Automated Trigger: When labor-to-revenue ratio exceeds risk-adjusted peer median by 15 points, the dashboard’s submission automatically triggers MAC review. The dashboard is not a passive transparency mechanism dependent on CMS reading 55 quarterly filings. It is an input to an automated enforcement trigger. CMS receives the data; the trigger fires algorithmically.

Standardized Template: CMS should prescribe a standardized template with mandatory line items to prevent OPOs from burying material facts in formatting. The template must require the encumbered/unencumbered breakdown on the first page, in a standardized table, before any narrative context.

4.4 IV.D. Board Independence and Direct Information Access

Requirement: At least 50% of OPO board members must have no current or prior (within 5 years) employment, contractual, or financial relationship with the OPO, its executives, its major vendors, or any tissue processing entity with which the OPO does business.

Additionally, each board member must annually certify in writing to CMS that they have received substantive operational information directly from at least one source other than the CEO or executive leadership. Qualifying sources include: participation in a clinical ride-along with an organ recovery coordinator; direct review of unedited quality data, cost reports, or personnel turnover reports; or an in-person meeting with frontline clinical staff conducted without executive presence.

Rationale: Board capture is not caused by a lack of data. It is caused by informational monopoly—the board receives all information through the CEO’s curation filter. Structural independence (composition requirements) plus informational independence (direct access certification) creates a governance architecture where independent directors can verify organizational reality through direct observation rather than executive narrative. Neither alone is sufficient. A structurally independent board member who receives all information through the CEO’s filter is functionally captured.

4.5 IV.E. Approval Chain Latency (ACL)

This metric formalizes the Performance Layer’s behavioral fingerprint: clinical recommendations that contradict executive preferences take longer to approve than those that align.

$$\text{ACL}_i = \frac{\text{Median}(t_{\text{approval}} \mid \text{contradict})}{\text{Median}(t_{\text{approval}} \mid \text{align})}$$

Where t_{approval} is the time from clinical recommendation submission to administrative approval, partitioned by whether the recommendation aligns with or contradicts executive-preferred outcomes (e.g., continuing vs. terminating donor management, accepting vs. declining a marginal organ).

- $\text{ACL} \approx 1.0$: approvals take the same time regardless of alignment. The decision process is clinically driven.
- $\text{ACL} \gg 1.0$: contradicting recommendations face systematic delays. The decision process is politically filtered.

Case evidence documents approval gates where clinical recommendations required executive sign-off that was withheld or delayed when the recommendation conflicted with operational preferences.

Measurement: Computable from event-level decision logs required under the Decision-Process Telemetry provision (Section IX). The classification of “align” vs. “contradict” is determined retrospectively by comparing the recommendation against the organization’s historical pattern of donor acceptance/declination decisions.

Threshold: $\text{ACL} > 2.0$ (contradicting recommendations take more than twice as long to approve) triggers governance review. $\text{ACL} > 3.0$ triggers QAPI noncompliance finding under §486.348.

4.6 IV.F. Variance Asymmetry Detector (VAD)

Terminal Legibility—the encryption of D_{true} through AI-generated compliance artifacts—is the most sophisticated threat to governance measurement. The counter is the Variance Asymmetry Detector: measuring the divergence between controllable narrative outputs and uncontrollable physics-bound exhaust.

$$\text{VAD}_i = \frac{\text{Var}(\text{Narrative}_i)}{\text{Var}(\text{Exhaust}_i)}$$

Where:

- Narrative_i = controllable artifacts: financial reports, QAPI narratives, board presentations, strategic plans. These can be smoothed, curated, and AI-generated to arbitrary precision.

- Exhaust_i = physics-bound signals: organ yield ratios (binary: organ on table or not), cold ischemic time distributions, LinkedIn clinical turnover velocity, state unemployment filings, transponder-trackable transport logistics, hospital referral response times.

When $\text{VAD} \ll 1.0$, the organization's narrative is suspiciously smooth relative to its operational reality. Smooth financials with chaotic operational indicators are the signature of encryption running. An organization genuinely performing well has correlated narrative and exhaust variance—good reports reflect good operations. An organization encrypting D_{true} has divergent variance—smooth reports masking volatile operations.

Implementation: The narrative inputs are already collected (cost reports, QAPI reports, board minutes). The exhaust inputs require integration of external data sources during recertification survey: LinkedIn turnover velocity for clinical roles (publicly scrapeable), state unemployment filings (obtainable via data-sharing agreement), organ yield ratios and ischemic time data (already collected by UNOS/OPTN). The VAD is computed during each survey cycle, not quarterly, because it requires manual integration of external data.

Threshold: $\text{VAD} < 0.3$ (narrative variance is less than 30% of exhaust variance) flags the organization for enhanced review. This threshold should be calibrated during the shadow period by computing VAD for all 55 OPOs and identifying the distribution.

5 V. Allocation Sequence Integrity

5.1 V.A. AOOS Enforcement Without Safe Harbor

Out-of-sequence kidney allocation increased from $< 3\%$ pre-2021 to nearly 20% by end of 2023. Five OPOs accounted for 29% of all out-of-sequence kidneys. Over 1,200 people died after being skipped on a waiting list in the past five years.

Critical design choice: No percentage threshold. Setting any threshold (15%, 20%, any number) implicitly legalizes the practice up to that threshold. Under Goodhart’s Law, OPOs will algorithmically optimize bypass volume to remain just below the enforcement line. The framework eliminates safe harbors entirely.

Requirement: Every out-of-sequence allocation must be documented with: (a) standardized reason code (not free-text narrative); (b) objective cold ischemic time data at the point the bypass decision was made; (c) documented, timestamped declinations from the top 5 match-run candidates with specific anatomic/biopsy reasons; (d) demographic characteristics of the highest-ranked skipped patient; (e) outcome tracking (was the organ transplanted and the 6-month graft survival).

5.1.1 Financial Consequence: SAC Reimbursement Review

Any out-of-sequence allocation that lacks documented objective justification (verified cold ischemic time data demonstrating imminent organ failure, or documented declinations from top-5 match candidates) is flagged for MAC reimbursement review of the associated Standard Acquisition Charge. The MAC reviews and may deny or reduce the SAC for organs placed without adequate justification.

This is self-executing. OPOs that bypass the list for logistical convenience or local favoritism face \$40,000+ per-organ financial risk. OPOs that bypass the list for genuine clinical necessity (a kidney at 18 hours of cold time with five documented declinations) are reimbursed normally. The enforcement mechanism is financial consequence, not bureaucratic monitoring.

5.1.2 Gaming Resistance

Anticipated gaming: coordinators instructed to select “Imminent Ischemic Failure” from a drop-down menu for every bypass. Counter: the reason code must be accompanied by objective CIT data (organ monitoring telemetry or documented time-stamps). A coordinator cannot claim ischemic failure at 6 hours of cold time when the median cold ischemic time for deceased-donor kidneys is 16–18 hours. The objective data falsifies the subjective code.

Anticipated gaming level 2: deliberately running a slow match run to let CIT accumulate, then declaring ischemic failure. Counter: match-run initiation time is logged by UNOS systems. The delta between organ recovery completion and match-run initiation is independently measurable. Systematic delays (match run initiated > 2 hours after recovery) are flagged and reported.

5.2 V.B. Symmetrical Transplant Center Accountability

Transplant centers that systematically decline medically complex organs—organs subsequently transplanted successfully elsewhere—should face accountability. If a transplant center’s organ declination rate significantly exceeds peer benchmarks for comparable organ quality, and declined organs achieve $> 80\%$ graft survival at accepting centers, this should be reported to the OPTN and considered in the transplant center’s Conditions of Participation review.

5.2.1 Declination Futility Index (DFI)

$$DFI_i = \frac{\text{Organs declined by center } i \text{ that were successfully transplanted elsewhere}}{\text{Total organs offered to center } i}$$

DFI quantifies how often a center declines organs that other centers successfully use. A high DFI indicates either excessively conservative selection criteria or capacity management that prioritizes logistical convenience over patient access. $DFI > 0.15$ (more than 15% of offered organs declined by the center were successfully used elsewhere) triggers OPTN review.

5.3 V.C. Medically Complex Organ Credit

Medically complex organs (DCD donors, elevated KDPI scores, HIV+/HCV+ organs) require greater clinical effort, higher logistical cost, and more sophisticated family authorization conversations. Under current outcome measures, a medically complex organ transplant counts identically to a standard transplant. This creates an incentive to pursue the easiest donors rather than maximize total transplantable organs.

Requirement: CMS should apply a $1.5\times$ multiplier in the outcome measures for each organ from a medically complex donor (as newly defined at §486.302) that is successfully transplanted. A DCD kidney transplanted successfully counts as 1.5 organs in the donation rate and transplant rate calculations.

$$O_{i,\text{weighted}} = O_{i,\text{standard}} + 1.5 \times O_{i,\text{complex}}$$

The weighted output is used in REI computation. This directly incentivizes the clinical investment, family authorization effort, and transplant center partnership development required to recover and place the approximately 12,000 organs recovered but discarded annually.

6 VI. Tissue-Organ Financial Separation

Organ procurement operates on Medicare cost-pass-through reimbursement. Tissue recovery operates on market-priced revenue. When the two are financially intermingled, the incentive to maximize tissue revenue displaces the mission of maximizing organ recovery. One OPO reported spending \$392M on tissue processing versus \$22M on organ procurement while simultaneously achieving Tier 3 status.

Requirement 1 (Disclosure): OPO executives and board members must disclose all financial relationships with tissue processing entities, funeral homes, and medical examiner offices.

Requirement 2 (Prohibition): OPO executives and advisory board members are prohibited from holding equity interests in, receiving compensation from, or serving on the board of any tissue processing entity, regardless of tax-exempt status, or any entity that does business with the OPO.

Requirement 3 (Separate Reporting): Tissue-related revenue and expenses must be reported on separate cost report line items from organ acquisition costs. Shared overhead (facilities, IT infrastructure, HR, executive compensation) must be allocated using a standardized methodology prescribed by CMS, not by the OPO's internal cost accountants.

6.0.1 Zero-Contamination Principle (Long-Term)

For the most rigorous separation: any facility, software system, or employee used for both organ recovery and tissue recovery should have its shared cost allocated proportionally based on time-study data, not managerial discretion. In future rulemaking, CMS should consider requiring that Medicare-reimbursable assets and personnel be exclusively dedicated to organ recovery—if an asset is used for tissue recovery, its cost is disallowed from the Medicare cost report to the extent of tissue utilization.

7 VII. The Five-Level Graduated Intervention Ladder

The current system provides three outcomes: Tier 1 (retain DSA), Tier 2 (compete for DSA), Tier 3 (lose DSA). There is no intermediate intervention. This binary structure creates perverse incentives: merger arbitrage, narrative theater, and scapegoating. The replacement is a five-level ladder with graduated consequences.

7.1 Level 1: Full Certification

REI ≥ 1.0 , CPTO within $1.25\times$ risk-adjusted median, no governance flags. Status: unrestricted. 4-year recertification cycle. Minimal reporting.

7.2 Level 2: Monitoring

REI 0.85–0.99, OR CPTO $1.25\text{--}1.5\times$ median, OR minor governance concerns (single department turnover anomaly, single quarter financial reporting discrepancy). Status: enhanced quarterly reporting. CMS assigns a monitoring liaison. No operational restrictions. OPO develops voluntary improvement plan. Duration: up to 8 quarters. Return to Level 1 upon sustained improvement.

7.3 Level 3: Improvement Required

REI 0.70–0.84 for two consecutive years, OR CPTO $> 1.5\times$ median for two consecutive years, OR significant governance findings (captured accountability mechanisms, sustained personnel anomalies, executive compensation grossly disconnected from performance).

Status: Mandatory performance improvement plan approved by CMS. Quarterly reporting to CMS and board. CMS may require independent management consultant.

7.3.1 The Medicare Cost Disallowance Mechanism

CMS lacks statutory authority to freeze private employment contracts. CMS possesses absolute authority over what Medicare reimburses. When an OPO enters Level 3:

Executive compensation above the 75th percentile of clinical coordinator compensation at that OPO is designated as a strictly unallowable cost on the Medicare cost report until the OPO returns to Level 2 or above.

The OPO board retains full authority to pay its executives any amount it chooses—from tissue revenue, reserves, or private fundraising. Medicare simply stops subsidizing it. Zero Medicare dollars may be disbursed for executive overhead above the threshold; 100% of federal funds must be restricted to direct frontline clinical labor and organ transport.

Thermodynamic effect: The board must drain its own private, unrestricted reserves to maintain executive compensation during underperformance. This creates fiduciary pressure: the board will either improve operations (restoring Medicare reimbursement eligibility) or replace leadership (reducing the cash drain). A CEO earning \$709,000 annually from Medicare pass-through suddenly costs the organization \$709,000 from reserves—approximately \$59,000 per month of private cash burn. The incentive to sustain narrative theater inverts: improvement becomes the cheapest path to compensation restoration, not narrative management.

Duration: up to 12 quarters. Failure to improve triggers Level 4.

7.4 Level 4: Supervised Operation

REI < 0.70 for two consecutive years, OR failure to improve from Level 3 within 12 quarters, OR critical governance failure (documented fraud, evidence destruction, regulatory gaming).

Status: CMS appoints a temporary oversight manager with authority to approve major financial decisions, executive hires, and strategic commitments. DSA opened to competitive expressions of interest (not yet a formal bid process, but CMS begins identifying potential successor operators).

Clinical Workforce Protection: No involuntary separation of clinical workforce during supervised period without CMS approval. This protects the core wire during governance transition. Administrative and executive staff are not protected.

D&O Insurance Notification: Upon entry into Level 4, CMS publishes the organization's governance metrics. D&O insurance carriers, upon seeing public documentation of governance failures at a supervised entity, will re-price coverage or add exclusions. This creates an independent financial forcing function: the board's personal liability shield becomes more expensive precisely when the organization's governance is worst, aligning carrier incentives with reform.

Duration: up to 8 quarters. Return to Level 3 upon sustained improvement. Failure to improve triggers Level 5.

7.5 Level 5: Decertification and Transition

Failure to improve from Level 4 within 8 quarters, OR REI < 0.50 for any single year (emergency trigger), OR documented fraud of sufficient magnitude to warrant immediate action.

Status: CMS initiates competitive bid process for DSA. Incumbent OPO continues operating during transition (LAORA precedent: 112 days from announcement to successor selection). All clinical staff are offered employment by successor operator as condition of bid acceptance (the Clinical Workforce Retention Covenant). Administrative and executive staff are not guaranteed retention. Transition managed by CMS-appointed administrator.

7.5.1 The Clinical Workforce Retention Covenant

Any entity bidding to operate a DSA vacated by a decertified OPO must, as a condition of bid acceptance, offer employment to substantially all clinical staff of the predecessor OPO at comparable compensation and benefits. This protects:

- Hospital relationships: ORCs who have built trust with donor hospital staff over years carry institutional knowledge that cannot be rebuilt from scratch.
- Community trust: family services staff known to the community maintain the consent-rate relationships that drive organ recovery.
- Operational continuity: the clinical operation continues uninterrupted during governance transition.
- Employee welfare: the clinical workforce was never the source of governance dysfunction and should not bear the consequences of leadership failure.

8 VIII. Competitive Entry and Exit Architecture

8.1 VIII.A. New Entrant Certification Requirements

CMS solicits comments at Section III.B on “specific elements an entity should meet to become a newly certified OPO.” The following requirements ensure that new entrants have the operational, financial, and governance infrastructure to sustain organ recovery operations from day one:

- (a) Financial reserves: 12 months of projected operating expenses held in unrestricted, liquid assets, sufficient to fund full operation without Medicare reimbursement during the startup period before cost report reconciliation produces cash flow.
- (b) Transplant center partnerships: Binding agreements with transplant centers covering at least 60% of transplant volume in the target DSA, ensuring recovered organs have established placement pathways from day one.
- (c) Governance structure: Meeting all Conditions for Coverage including board independence, external ethics infrastructure, financial transparency dashboard, and personnel pattern monitoring requirements proposed in this framework.
- (d) Clinical staffing plan: Demonstrating the ability to provide 24/7 on-call coverage with appropriately licensed personnel meeting the requirements of proposed §486.326(e), with named individuals or binding recruitment agreements.
- (e) Workforce transition plan: Specifying how the applicant will offer employment to substantially all clinical staff of the predecessor OPO at comparable compensation (the Clinical Workforce Retention Covenant).
- (f) Technology infrastructure: Operational EMR, UNOS DonorNet access, real-time referral tracking capability, and decision-process telemetry logging meeting the requirements of Section IX of this framework.

8.2 VIII.B. DSA Boundary Flexibility Algorithm

The current system treats DSA boundaries as fixed geographic units. When a DSA is structurally non-viable (high REI but low AOC—the OPO is doing its job but the territory can’t sustain a standalone operation), the framework should permit boundary redesignation.

8.2.1 Algorithm

For each DSA_i where $AOC_i < \text{Floor}$ AND $REI_i \geq 1.0$:

1. Identify adjacent DSAs $\{j\}$ where $REI_j \geq 1.1$
2. For each adjacent DSA_j , compute marginal REI impact:

$$\Delta REI_j = REI(O_j + O_i, E_j + E_i) - REI_j$$

3. If $\Delta REI_j > -0.05$ (absorbing OPO i ’s territory does not degrade OPO j ’s performance by more than 5%): $\rightarrow DSA_i$ is a candidate for merger into DSA_j
4. Select $j^* = \arg \max(REI_j)$ among qualifying candidates
5. Initiate boundary redesignation proceeding

This is the mechanism that makes AOC actionable. When a territory cannot sustain standalone operations, the answer is not to punish the OPO—it is to redesignate the territory to an adjacent high-performer that can absorb it without degradation. The clinical workforce in the redesignated territory transfers to the absorbing OPO under the Workforce Retention Covenant.

8.3 VIII.C. Anti-Consolidation Threshold

To prevent a single entity from accumulating monopoly power at national scale, no OPO should operate DSAs accounting for more than 15% of national estimated biological potential (eligible deaths). This prevents the replacement of 55 small monopolies with 3–4 large ones.

9 IX. Technology Integration and AI Governance

9.1 IX.A. Real-Time Referral Tracking

Every hospital referral timestamped from notification to OPO response, with response time benchmarked against the national distribution and published quarterly. This measures the first link in the clinical chain—how quickly the OPO responds when a hospital calls.

$$\text{ResponseTime}_i = t(\text{OPO_first_contact}) - t(\text{hospital_notification})$$

Published as distribution (median, 90th percentile, max) per OPO per quarter alongside tier determinations.

9.2 IX.B. Authorization Tracking by Demographics

Consent/authorization rates tracked by donor demographics (age, race/ethnicity, cause of death, hospital type), benchmarked against DSA-specific expected rates from the risk adjustment model.

$$\text{ConsentGap}_{i,g} = \text{AuthRate}_{i,g} - E[\text{AuthRate}_{i,g} \mid X_i]$$

Where g indexes demographic group. An OPO whose consent rate for a specific population is significantly below the DSA-adjusted expectation has a measurable problem that can be targeted with culturally competent outreach. The gap is published to make disparities visible and actionable.

9.3 IX.C. Decision-Process Telemetry

As OPOs adopt AI-assisted clinical tools, the framework requires event-level decision logging:

Every clinical decision point where a human overrides an algorithmic or protocol recommendation must be logged with: the decision, the algorithmic recommendation, the human’s documented reasoning, the clinical outcome (tracked at 30 days, 90 days, 1 year), and the timestamp.

9.3.1 The Zero-Override Principle

A zero override rate is a surveillance finding, not a compliance success. An OPO where no clinician ever overrides a protocol recommendation is an OPO where clinicians have stopped exercising professional judgment. The healthy signal is a calibrated override rate—overrides at a frequency consistent with genuine professional judgment, with documented reasoning, and with outcomes that can be retrospectively evaluated.

Both extremes trigger review:

- Override rate $\approx 0\%$: Automation bias. Clinicians are following algorithms uncritically.
- Override rate systematically suppressed (overrides present but declining to zero over quarters): Compliance theater. The organization is training clinicians to stop documenting legitimate overrides.
- Override rate within calibrated range (5–20%, varying by decision type): Healthy professional judgment.

9.4 IX.D. Terminal Legibility Detection

Terminal Legibility is the encryption of D_{true} through AI-generated compliance artifacts so flawless that external enforcement cannot distinguish appearance from reality. The counter is the Variance Asymmetry Detector (Section IV.F) combined with decision-process telemetry.

Event-level data is much harder to fabricate than narrative compliance reports because it must be internally consistent across thousands of individual logged decisions. An AI system generating fake decision logs must maintain consistency across every entry, every timestamp, every participant’s schedule, every organ’s documented ischemic time. The thermodynamic cost of fabricating event-level data at scale exceeds the cost of simply operating honestly. This is the design intent.

9.4.1 The AI Standard of Care Paradox

When an OPO deploys AI compliance tools that create a perfect, timestamped audit trail, it has automated away the epistemic friction that previously provided legal cover for the board. “We couldn’t have known” becomes provably false when the organization’s own AI system logged the data that would have informed better decisions. Terminal Legibility does not just encrypt D for external observers—it creates discoverable evidence that the organization had the means to know and chose not to act. The encryption is also the prosecution exhibit.

This paradox is a feature, not a bug. It means that as AI adoption increases across OPOs, the legal standard of care for board oversight increases in lockstep. Organizations cannot claim ignorance when they operate systems specifically designed to eliminate ignorance.

10 X. Risk Adjustment Model: Full Technical Specification

10.1 X.A. Estimation Methodology

The model is a negative binomial regression estimated on the national cross-section of all 55 DSAs using 3–5 years of pooled data.

$$O_{it} \sim \text{NegBin}(\mu_{it}, \alpha)$$

$$\ln(\mu_{it}) = X_{it}\beta + \ln(\text{Pop}_{it}) \quad [\text{offset for population}]$$

The negative binomial is preferred over Poisson because organ counts exhibit overdispersion (variance > mean) driven by unmeasured DSA heterogeneity. The dispersion parameter α is estimated from the data.

10.2 X.B. Variable Selection

Variable	Source	Rationale	OPO Control?
Eligible deaths per 100K	CDC MCODE	Primary denominator input	No
Population density	Census	Transport feasibility, ischemic time	No
Trauma center density	AHA Survey	Referral infrastructure quality	No
Registered donor rate	Donate Life America	Cultural baseline (state level)	Marginal (slow-moving)
Geographic dispersion	GIS computation	Transport cost/time	No
Median household income	Census/ACS	Healthcare access proxy	No
Transplant center count	SRTR	Placement capacity	No

10.3 X.C. Validation Protocol

Step 1 (Backtesting): Estimate the model on 2018–2022 data. Compute predicted REI for all 55 OPOs for 2023. Compare predicted Tier classification to actual Tier classification. Identify discrepancies—OPOs that are Tier 3 under actual data but $\text{REI} > 0.85$ under the model (structurally disadvantaged DSAs being unfairly penalized by the current system) and OPOs that are Tier 1 but $\text{REI} < 0.85$ (structurally advantaged DSAs masking mediocre performance).

Step 2 (Out-of-sample): Estimate on 2018–2021, predict 2022–2023, evaluate RMSE and calibration. The model should be well-calibrated: among OPOs predicted to achieve $\text{REI} = 1.0$, approximately 50% should exceed 1.0 and 50% should fall below.

Step 3 (Sensitivity): Compute the change in each OPO's REI classification when each variable is excluded. If any single variable changes more than 5 OPOs' classifications, that variable requires additional scrutiny (high leverage).

Step 4 (Publication): Publish the model specification, coefficients, standard errors, and diagnostic statistics. Invite public comment. Any OPO should be able to replicate its own REI computation from published data.

10.4 X.D. Shadow Period Design

Phase	Timeline	Actions
Year 1 (Rulemaking)	Publish proposed model	Notice and comment. Solicit input on variable selection, threshold structure, and the demographic adjustment question.
Year 2 (Parallel Operation)	Compute REI for all 55 OPOs	Publish alongside tier determinations with no regulatory consequences. Identify discrepancies. Refine model.
Year 3 (Partial Integration)	REI informs graduated ladder	Secondary factor. High REI + Tier 2/3 → benefit of doubt. Low REI + Tier 1 → enhanced review.
Year 4+ (Full Integration)	REI becomes primary metric	Tier system retired. Coefficients updated each 4-year cycle.

11 XI. Gaming Resistance Architecture

Every metric in this framework will be optimized against. This section documents the anticipated gaming strategy for each metric, the counter-mechanism, and the residual vulnerability.

Metric	Gaming Strategy	Counter-Mechanism	Residual Risk
REI	Inflate eligible deaths denominator	CDC MCOD data, national definition. OPO has no input.	Low
REI #2	Recover marginal donors to inflate numerator	REI uses organs transplanted, not recovered. Recovering discarded organs does not help.	Low
CPTO	Shift costs to affiliates	Cross-validate cost report vs. 990. $\Delta > 10\%$ triggers audit.	Medium
CLE	Reclassify admin as clinical FTE	$> 50\%$ physical deployment criterion. Managing others = admin regardless of license.	Low
Dashboard	Bury facts in formatting	Standardized template. Automated trigger at 15pt LRR deviation.	Medium
Ethics hotline	Informally discourage use	WTR detects downstream retaliation. OIG routing prevents suppression.	Medium
Personnel	Reclassify involuntary as voluntary	Cross-validate vs. unemployment filings.	Low
AOOS	Select “ischemic failure” code for all bypasses	Objective CIT data required. 6hr CIT $\neq$ ischemic failure.	Medium
Tissue sep.	Create arms-length entities	UBO cross-match against vendor payments.	Medium
Exec comp	Defer below reporting threshold	Total comp including deferred and in-kind, per 990 Part VII.	Low

Metric	Gaming Strategy	Counter-Mechanism	Residual Risk
Board indep.	Appoint technically independent but captive directors	Direct information access certification. Audit during survey.	High
ACL	Approve everything quickly, override later	Track post-approval reversal rate. High reversal = sham approval.	Medium
VAD	Inject noise into narratives to match exhaust variance	Noise injection degrades report quality, inviting scrutiny.	High

11.0.1 The Meta-Gaming Proposition

A sufficiently sophisticated executive will optimize simultaneously across all metrics, maintaining each just above threshold. The framework’s meta-gaming resistance relies on a single proposition:

When the cheapest way to appear compliant across all metrics simultaneously is to actually run the organization well, the Performance Layer’s business model collapses.

The cost of gaming N independent metrics scales multiplicatively. Gaming one metric is cheap. Gaming 12 simultaneously—each with different data sources, different verification mechanisms, and different enforcement bodies—is more expensive than genuine compliance. This is the design intent. No single metric is sufficient. The system of metrics is the defense.

12 XII. Implementation and Transition Plan

12.1 XII.A. Four-Year Phased Transition

Phase	Timeline	Actions	Regulatory Vehicle
1	Year 1 (2027)	Publish proposed OPPGAS rule. Implement Proposals 1–4 (dashboard, ethics, personnel, efficiency) as CoC requirements.	Final rule CMS-3409-P (immediate provisions) + new NPRM for OPPGAS.
2	Year 2 (2028)	Implement Proposals 5–6 (AOOS enforcement, tissue separation). Begin REI shadow computation. Compute VAD for all 55 OPOs.	Final OPPGAS rule (Phase 1 provisions).
3	Year 3 (2029)	REI published alongside tiers. Graduated ladder operational (Levels 1–3). Competitive entry requirements codified. DSA boundary flexibility pilot.	Final OPPGAS rule (Phase 2 provisions).
4	Year 4+ (2030+)	Full framework operational. REI replaces raw tiers. Levels 4–5 activated. Boundary redesignation available. Coefficient updates.	Steady-state operations.

12.2 XII.B. Cost Estimates

Component	Annual Cost (National)	Per OPO
Financial Dashboard	\$200K–\$400K	\$3,600–\$7,300
External Ethics Infrastructure	\$825K–\$1.65M	\$15,000–\$30,000
Personnel Pattern Monitoring	\$300K–\$500K	\$5,500–\$9,100

Component	Annual Cost (National)	Per OPO
Efficiency Benchmarks	\$100K–\$200K	\$1,800–\$3,600
AOOS Enforcement	\$500K–\$1M	\$9,100–\$18,200
Tissue Financial Separation	\$300K–\$600K	\$5,500–\$10,900
Board Independence	\$150K–\$300K	\$2,700–\$5,500
Decision-Process Telemetry	\$400K–\$800K	\$7,300–\$14,500
Risk Adjustment Model	\$200K–\$400K	\$3,600–\$7,300
TOTAL	\$3.0M–\$5.9M	\$54,000–\$106,000

Total estimated annual cost: approximately \$3.0–\$5.9 million nationally, distributed across 55 OPOs. This represents approximately 0.008–0.015% of total OPO industry revenue, or roughly \$54,000–\$106,000 per OPO annually. The per-OPO cost is less than two months of CEO compensation at the documented organization. The national cost is less than the annual compensation of five OPO CEOs.

12.3 XII.C. Data Source Inventory

Data	Source	Currently Collected?	Access
Organ counts	UNOS/OPTN	Yes	Federal
Eligible deaths	CDC MCOD	Yes	Public
Cost reports	CMS	Yes	Federal
Form 990	IRS	Yes	Public
Population/demographics	Census/ACS	Yes	Public
Trauma centers	AHA Annual Survey	Yes	Subscription
Transplant center data	SRTR	Yes	Public
Donor registration	Donate Life America	Yes	Public
Unemployment filings	State UI agencies	No (new DSA needed)	DSA agreement
Clinical turnover	LinkedIn / OPO report	Partial	Public + reported
Match run data	UNOS	Yes	Federal
Bond documents	EMMA/MSRB	Yes	Public

13 XIII. Mathematical Appendix

This appendix provides formal definitions for all quantitative metrics in the framework, suitable for implementation by CMS actuaries and data scientists.

13.1 A. Recovery Efficiency Index

$$\text{REI}_i = \frac{O_i}{\hat{\mu}_i}$$

$$\hat{\mu}_i = \exp(X_i \hat{\beta} + \ln(\text{Pop}_i))$$

$$\hat{\beta} = \arg \max_{\beta} \sum_i [O_i \ln(\mu_i) - \ln(O_i!) - \dots] \quad (\text{NB MLE})$$

$$\text{SE}(\ln(\text{REI}_i)) = \sqrt{\frac{1}{O_i} + \frac{1}{\hat{\alpha}}} \quad [\text{approximate}]$$

$$\text{REI}_{i,\text{lower}} = \exp(\ln(\text{REI}_i) - 1.645 \times \text{SE}(\ln(\text{REI}_i)))$$

13.2 B. Performance Layer Deficit Estimation

$$\hat{D}_i = (\text{LRR}_i - \widetilde{\text{LRR}}) \times \text{Rev}_i + (\text{ECE}_i - \widetilde{\text{ECE}}) \times O_i$$

Where $\widetilde{\text{LRR}} = \text{median}(\text{LRR})$ across all OPOs, $\widetilde{\text{ECE}} = \text{median}(\text{ECE})$ across all OPOs.

Interpretation:

- $\hat{D} > 0 \rightarrow$ excess labor + excess executive compensation
- $\hat{D} > \$5\text{M} \rightarrow$ significant performance layer
- $\hat{D} > \$10\text{M} \rightarrow$ severe governance dysfunction

13.3 C. Variance Asymmetry Index

$$\text{VAD}_i = \frac{\sigma^2(N_{i,t})}{\sigma^2(E_{i,t})} \quad \text{over } T \text{ quarters}$$

Where $N_{i,t}$ = normalized narrative metric (cost report ratios, QAPI scores, board satisfaction indices) and $E_{i,t}$ = normalized exhaust metric (organ yield, clinical turnover rate, CIT distributions, referral response times).

- $\text{VAD} < 0.3 \rightarrow$ flag for Terminal Legibility review
- $\text{VAD} \approx 1.0 \rightarrow$ healthy correlation (reports reflect reality)
- $\text{VAD} > 3.0 \rightarrow$ noisy reports, stable operations (unusual, investigate)

13.4 D. Approval Chain Latency

$$\text{ACL}_i = \frac{\text{Median}(\{t_k : \text{decision}_k \in \text{Contradict}\})}{\text{Median}(\{t_k : \text{decision}_k \in \text{Align}\})}$$

Where t_k = time from recommendation to approval for decision k .

Classification: Contradict if recommendation $\neq$ historical pattern; Align if recommendation = historical pattern.

- $\text{ACL} > 2.0 \rightarrow$ governance review
- $\text{ACL} > 3.0 \rightarrow$ QAPI noncompliance finding

13.5 E. Whistleblower-to-Termination Ratio

$$\text{WTR}_{i,d} = \frac{W_{i,d} / \mu_W}{S_{i,d} / \mu_S}$$

Where $W_{i,d}$ = complaints in department d within 12 months, $S_{i,d}$ = involuntary separations in department d within 6 months after complaint period, $\mu_W = E[\text{complaints}]$ given org size, $\mu_S = E[\text{separations}]$ given org size.

- $\text{WTR} > 2.0 \rightarrow$ automatic on-site CMS review
- $\text{WTR} > 3.0 \rightarrow$ simultaneous OIG referral

13.6 F. Declination Futility Index

$$\text{DFI}_i = \frac{|\{\text{organs declined by center } i, \text{ transplanted elsewhere}\}|}{|\{\text{organs offered to center } i\}|}$$

- $\text{DFI} > 0.15 \rightarrow$ OPTN review triggered

13.7 G. Weighted Output (Medically Complex Credit)

$$O_{i,w} = O_{i,\text{standard}} + 1.5 \times O_{i,\text{complex}}$$

$$\text{REI}_{i,w} = \frac{O_{i,w}}{E[O_i | X_i]}$$

Substitutes for O_i in all performance calculations.

13.8 H. Anti-Consolidation Constraint

For any entity $\mathcal{E}$ operating DSAs $S_1, S_2, \dots, S_k$:

$$\frac{\sum_{j=1}^k D(S_j)}{D_{\text{national}}} \leq 0.15$$

Where $D(S_j)$ = eligible deaths in DSA j , D_{national} = total national eligible deaths.

Violation $\rightarrow$ entity barred from bidding on additional DSAs.

13.9 I. DSA Boundary Redesignation Criterion

Eligible if: $REI_i \geq 1.0$ AND $AOC_i < \frac{1}{2N}$

Absorbing DSA:

$$j^* = \arg \max \{ REI_j : \text{adj}(j, i), REI_j \geq 1.1, \Delta REI_j > -0.05 \}$$

$$\Delta REI_j = \frac{O_j + O_i}{E_j + E_i} - \frac{O_j}{E_j}$$

13.10 J. Medicare Cost Disallowance (Level 3)

$$\text{Allowable_exec}_i = \min(\text{ExecComp}_i, P_{75}(\text{CoordComp}_i))$$

$$\text{Disallowed}_i = \text{ExecComp}_i - \text{Allowable_exec}_i$$

Where $P_{75}(\text{CoordComp}_i)$ = 75th percentile of clinical coordinator compensation at OPO i .

Disallowed amount excluded from Medicare cost report. Board may fund from tissue revenue / reserves.

14 XIV. Conclusion

This document presents the complete, mathematically specified framework for organ procurement organization oversight. It integrates outcome measurement (REI, AOC), process efficiency (CPTO, CLE, AOR, ECE), governance quality (information flow integrity, personnel patterns, financial transparency, board independence, approval chain latency, variance asymmetry), allocation integrity (AOOS enforcement, symmetrical accountability, medically complex organ credit), structural separation (tissue-organ financial firewall), graduated enforcement (five-level ladder with Medicare cost disallowance), competitive architecture (new entrant requirements, DSA boundary flexibility, anti-consolidation constraints), and technology governance (real-time tracking, decision-process telemetry, Terminal Legibility detection).

The framework’s common design principle is architectural signal integrity: creating channels through which honest signals can travel without being intercepted, filtered, or suppressed by the subjects of oversight. The Performance Layer survives because it controls the channels. This framework creates channels it cannot control.

Some provisions are implementable within the current rulemaking cycle. Others require future rulemaking, data infrastructure development, or political will that may not yet exist. The framework is presented complete—including provisions that CMS may not adopt in 2026—because the regulatory architecture for the next decade should be designed now, even if it is built incrementally. The document that CMS files as “noted” today becomes the blueprint the next generation of regulators pulls off the shelf when the political will catches up to the math.

The clinical workforce at every OPO was never the problem. The coordinators who answer calls at 3 AM, the recovery teams, the family services staff, the surgical teams who recover organs under impossible time pressure—they operate at the core wire of organ procurement. Everything above them is scaffolding. This framework asks whether the scaffolding is load-bearing or parasitic, and builds the measurement architecture that makes the answer empirically undeniable.

Respectfully submitted.

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Related submissions on this docket: AORTA Framework (Comment 1); GOLA Theory (Comment 2, 43 pages); The Core Wire (Comment 3); The Performance Layer (Comment 4); Forensic Assessment (Comment 5, 20 pages); Structural Policy Proposals (Comment 6, 14 pages); Supplemental Errata and Additional Proposals (Comment 7).